

Faculty vitae: Naveen Dandu

1. Education

Dec 2016 Ph.D. in Chemistry, North Dakota State University, North Dakota, USA
May 2013 M.S. in Computer Science, North Dakota State University, North Dakota, USA
Jun 2007 M.S. in Chemistry, University of Hyderabad, Hyderabad, TG, INDIA
May 2003 B.S. in Chemical Technology, Loyola Academy, Hyderabad, TG, INDIA

2. Academic Experience

02/2022 - present Research Assistant Professor, Chemical Engineering, University of Illinois at Chicago, Illinois, USA
02/2019 - 02/2022 Postdoctoral Appointee, Materials Science Division, Argonne National Lab, Lemont, Illinois, USA
02/2017 - 02/2019 Postdoctoral Researcher, Department of Chemistry, University of Nevada, Reno, Nevada, USA
06/2014 - 08/2014 Summer Intern, Los Alamos National Lab, Los Alamos, New Mexico, USA
06/2013 - 08/2013 Summer Intern, Los Alamos National Lab, Los Alamos, New Mexico, USA
12/2010 - 01/2012 Research Assistant, Center for Biopharma. R&D, Fargo, North Dakota, USA
05/2003 - 05/2005 Research Assistant, Indian Institute of Chemical Technology, Hyderabad, TG, INDIA

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers
Association of Environmental Engineering and Science Professors

6. Selected Honors and awards

Not applicable

7. Selective Service Activities

- Serving as peer reviewer for Computational Materials Science Journal at Elsevier Publishers, Polymers for Advanced Technologies, ACS Catalysis, The Journal of Physical Chemistry,

iScience, Scientific Reports – open access, and Journal of Chemical Information and Modeling. (so far reviewed 40 articles)

- Teaching graduate and undergraduate level course titled "molecular and macromolecular engineering" (01/2023 to 05/2023).
- Served as a judge at 2020 National Science Bowl (01/2020)
- Served as a judge at Graduate Student Association Spring Awards – Research Grant Competition, University of Nevada, Reno, NV (03/2017)
- Volunteered in North Dakota Science Olympiad Program as an assistant organizer in forensic science division, North Dakota State University, Fargo, ND (04/2016)

8. Selected Publications Last 5 Years

Naveen K. Dandu, Alex Taekyung Lee, Sergio Ulloa, Larry Curtiss, Saw Wai Hla and Anh T. Ngo. Exploring the Coexistence of Spin States in $[\text{Fe}(\text{tpy-Ph})_2]^{2+}$ Complexes on Au(111) Using DFT Calculations. *J. Phys. Chem. A*, 129, 40, 9185–9194 (2025).

Mohammed Lemaalem, Selva Chandrasekaran Selvaraj, Ilias Papailias, Naveen K. Dandu, Arash Namaeighasemi, Larry A. Curtiss, Amin Salehi-Khojin and Anh T. Ngo. Interactive multiscale modeling to bridge atomic properties and electrochemical performance in Li-CO₂ battery design. *Applied Energy*, Volume 401, Part B (2025).

Ajayi, T. M.; Shirato, N.; Rojas, T.; Wieghold, S.; Cheng, X.; Latt, K. Z.; Trainer, D. J.; Dandu, N. K.; Li, Y.; Premarathna, S.; Sarkar, S.; Rosenmann, D.; Liu, Y.; Kyritsakas, N.; Wang, S.; Masson, E.; Rose, V.; Li, X.; Ngo, A. T.; Hla, S.-W. Characterization of Just One Atom Using Synchrotron X-Rays. *Nature* **2023**, 618 (7963), 69–73. <https://doi.org/10.1038/s41586-023-06011-w>.

Dandu, N. K.; Ward, L.; Assary, R. S.; Redfern, P. C.; Curtiss, L. A. Accurate Prediction of Adiabatic Ionization Potentials of Organic Molecules Using Quantum Chemistry Assisted Machine Learning. *The Journal of Physical Chemistry A* **2023**, 127 (28), 5914–5920. <https://doi.org/10.1021/acs.jpca.3c00823>.

Dandu, N. K.; Ward, L.; Assary, R. S.; Redfern, P. C.; Curtiss, L. A. Accurate Prediction of Adiabatic Ionization Potentials of Organic Molecules Using Quantum Chemistry Assisted Machine Learning. *The Journal of Physical Chemistry A* **2020**, 124 (28), 5804–5811. <https://doi.org/10.1021/acs.jpca.3c00823>.

Dandu, N. K.; Assary, R. S.; Redfern, R. C.; Ward, L.; Foster, I. and Curtiss, L. A. Improving the Accuracy of Composite Methods: A G4MP2 Method with G4-like Accuracy and Implications for Machine Learning. *J. Phys. Chem. A* **2022**, 126, 27, 4528–4536. <https://pubs.acs.org/doi/abs/10.1021/acs.jpca.0c01777>

Hemmat, Z.; Ahmadiparidari, A.; Wang, S.; Kumar, K.; Zepeda, M.; Zhang, C.; Dandu, N.; Rastegar, S.; Majidi, L.; Jaradat, A.; Ngo, A.; Thornton, K.; Curtiss, L. A.; Cabana, J.; Huang, Z.; Salehi-Khojin, A. Unprecedented Multifunctionality in 1D Nb_{1-x}Ta_xS₃ Transition Metal

Trichalcogenide Alloy. *Advanced Functional Materials* **2022**, 32 (34).
<https://doi.org/10.1002/adfm.202205214>.

Ward, L.; Dandu, N. K.; Blaiszik, B.; Narayanan, B.; Assary, R.; Redfern, P.C.; Foster, I. and Curtiss, L. A. Graph-Based Approaches for Predicting Solvation Energy in Multiple Solvents: Open Datasets and Machine Learning Models. *J. Phys. Chem. A* **2021**, 125, 27, 5990–5998.
<https://doi.org/10.1021/acs.jpca.1c01960>

Ward, L.; Sivaraman, G.; Pauloski, J. G.; Babuji, Y.; Chard, R.; Dandu, N. K.; Redfern, P.C.; Assary, R.; Chard, K.; Curtiss, L. A.; Thakur, R. and Foster, I. Colmena: Scalable Machine-Learning-Based Steering of Ensemble Simulations for High Performance Computing. 2021 IEEE/ACM Workshop on Machine Learning in High Performance Computing Environments (MLHPC), St. Louis, MO, USA, **2021**, pp. 9-20. doi: 10.1109/MLHPC54614.2021.00007.
<https://ieeexplore.ieee.org/abstract/document/9653177>

9. Selected Presentations

Year	Title of the talk	Conference Details
2024	Excited State Calculations of Lanthanum and Europium based Critical Materials (Invited talk)	<i>Critical Materials workshop, Athens, Ohio</i>
2024	A Time-Dependent DFT: Concepts and Applications (Invited talk)	<i>INDO-POLISH international workshop</i>
2019	Accurately Predicting Energies of Organic Molecules using Quantum Chemistry Informed Machine Learning	<i>Telluride Scientific Research Center Telluride, CO</i>
2018	Accurate computation of the spin state energetics of transition metal complexes. Role of exchange-correlation functional, scalar-relativistic and solvent effects	<i>ACS conference, Boston, MA</i>
2018	Development of descriptors for methane-to-methanol conversion and side-reactions on zeolite-supported tri-metal tri-oxo catalysts	<i>ACS conference, New Orleans, LA</i>
2018	Performance of density functional theory for predicting hydrogenation and dimerization of ethylene using transition metal hydride catalysts supported on zeolites	<i>ACS conference, New Orleans, LA</i>

10. Professional Development Activities

Not applicable